Advanced Concepts

Helmut Simonis

Constraint Based Production Scheduling

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Key Points

- We present some more advanced concepts in scheduling
- These occur in more specialized problem areas
- Typically require more work on modelling
- Solver support may be limited

Sequence Dependent Setup-Time ✓

- Our usual disjunctive resource model assumes that we can change easily from one task to the next
- There might be a cleaning/setup time required
  - This is part of the fixed duration part of a processStep description
- In some cases it is more complex
  - On some machines there is a setup-time required which depends on both the previous and the next product
  - This time varies significantly between product combinations
  - Typically, the time depends on some properties of the products
- The setup time is non-productive, and should be avoided when possible

Computed Setup-Time Matrix

Product | Setup Matrix ×

	Row	pt1	pt10	pt11	pt12	pt13	pt14	pt15	pt16	pt17	pt18	pt19	pt2	pt20	pt21	pt22	pt23	pt24	pt25	pt26	pt27	pt28
1	pt1	0	90	90	90	90	90	90	90	90	90	90	30	90	90	90	90	90	90	90	90	90
2	pt10	90	0	30	60	90	60	60	30	60	60	60	90	90	30	60	90	60	90	60	60	60
3	pt11	90	30	0	30	90	60	60	30	60	60	60	90	90	60	60	90	60	90	60	60	60
4	pt12	90	60	30	0	60	90	60	60	60	60	60	90	90	90	60	60	30	60	90	90	30
5	pt13	90	90	90	60	0	90	90	60	90	90	60	90	90	90	90	60	60	60	90	60	30
6	pt14	90	60	60	90	90	0	90	60	90	60	90	90	60	30	90	30	90	60	30	30	90
7	pt15	90	60	60	60	90	90	0	60	60	30	60	90	90	90	30	90	60	60	60	90	60
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11	pt19	90	60	60	60	60	90	60	30	30	60	0	90	90	90	30	90	60	90	90	90	30
12	pt2	30	90	90	90	90	90	90	90	90	90	90	0	90	90	90	90	90	90	90	90	90
13	pt20	90	90	90	90	90	60	90	90	90	60	90	90	0	90	90	60	90	90	90	60	90
14	pt21	90	30	60	90	90	30	90	60	90	90	90	90	90	0	90	60	90	60	30	60	90
15	pt22	90	60	60	60	90	90	30	60	30	60	30	90	90	90	0	90	60	60	60	90	60
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17	pt24	90	60	60	30	60	90	60	60	30	60	60	90	90	90	60	60	0	60	90	90	30
18	pt25	90	90	90	60	60	60	60	90	90	90	90	90	90	60	60	30	60	0	30	90	60
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21	pt28	90	60	60	30	30	90	60	30	60	60	30	90	90	90	60	60	30	60	90	90	0
22	pt29	90	60	60	90	90	30	90	60	90	60	90	90	60	60	90	60	90	90	60	30	90
23	pt3	90	30	30	60	90	30	60	30	60	30	60	90	60	60	60	60	60	90	60	30	60
24	pt30	90	90	90	90	90	60	90	90	60	60	60	90	60	90	60	60	90	90	90	60	90
25	pt31	90	60	60	90	90	60	90	60	90	90	90	90	90	60	90	90	90	90	60	60	90
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29	pt35	90	90	90	60	60	90	60	90	90	90	90	90	90	90	60	60	60	30	60	90	60
30	pt36	90	30	60	60	90	90	30	60	60	30	60	90	90	60	60	90	60	90	90	90	60
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38	pt43	90	60	60	60	90	90	60	60	30	60	30	90	90	90	30	90	60	90	90	90	60
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41	pt46	90	60	60	30	60	90	60	60	60	60	60	90	90	90	60	60	30	60	90	90	30
42	pt47	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90

Row	pt29	pt3	pt30	pt31	pt32	pt33	pt34	pt35	pt36	pt37	pt38	pt39	pt4	pt40	pt41	pt42	pt43	pt44	pt45	pt46	pt47
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pt42	60	60	90	60	90	90	90	90	90	90	90	90	90	60	90	0	90	60	90	90	90
pt43	90	60	60	60	90	60	30	90	60	60	60	90	30	60	90	90	0	90	60	30	90
pt44	30	60	90	60	60	90	90	60	60	90	90	60	90	90	90	60	90	0	90	90	90
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pt46	90	60	90	60	90	90	30	60	60	60	90	60	60	90	90	90	30	90	60	0	90
pt47	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	0

Row	pt48	pt49	pt5	pt50	pt51	pt52	pt53	pt54	pt55	pt56	pt57	pt58	pt59	pt6	pt60	pt61	pt62	pt63	pt64	pt65
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pt11	90	90	60	90	90	90	90	90	90	90	90	90	90	60	90	90	90	90	90	90
pt12	90	90	30	90	90	90	90	90	90	90	90	90	90	30	90	90	90	90	90	90
pt13	90	90	60	90	90	90	90	90	90	90	90	90	90	60	90	90	90	90	90	90
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pt15	90	90	60	90	90	90	90	90	90	90	90	90	90	60	90	90	90	90	90	90
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pt17	90	90	60	90	90	90	90	90	90	90	90	90	90	60	90	90	90	90	90	90
pt18	90	90	60	90	90	90	90	90	90	90	90	90	90	30	90	90	90	90	90	90
pt19	90	90	60	90	90	90	90	90	90	90	90	90	90	60	90	90	90	90	90	90
pt2	60	90	90	60	30	60	30	90	30	60	30	90	60	90	30	60	90	60	60	30
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pt28	90	90	30	90	90	90	90	90	90	90	90	90	90	30	90	90	90	90	90	90
pt29	90	90	90	90	90	90	90	90	90	90	90	90	90	60	90	90	90	90	90	90
pt3	90	90	60	90	90	90	90	90	90	90	90	90	90	30	90	90	90	90	90	90
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pt35	90	90	60	90	90	90	90	90	90	90	90	90	90	60	90	90	90	90	90	90
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pt44	90	90	60	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90
pt45	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90
pt46	90	90	30	90	90	90	90	90	90	90	90	90	90	30	90	90	90	90	90	90
pt47	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90	90

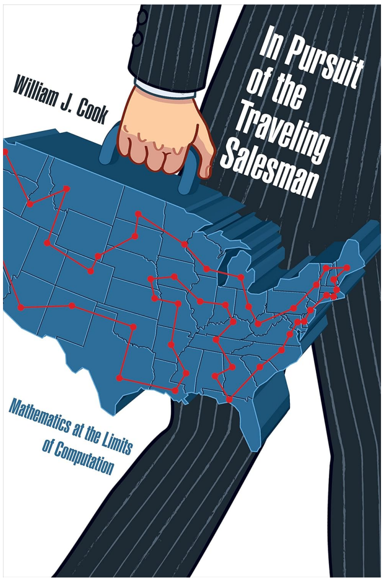
- This needs to be computed from first principles, not maintained by hand!
- Available as input data in JSON format

Relation to TSP

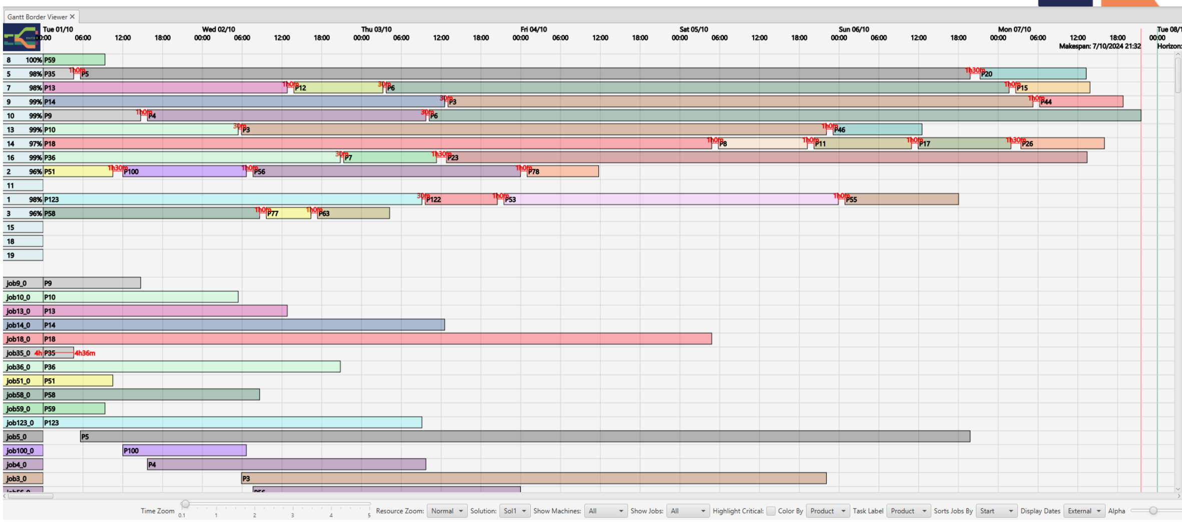
- Computing the optimal sequence of setup times is a variant of the *Travelling Salesman Problem (TSP)*
- Another of the classical hard combinatorial problems
- Due to the structure of the data, setup-time problems often are simpler to solve
  - Changing between very similar products needs no setup-time
  - Using a simple rule about product compatibility produces best results
  - Example: dark-chocolate → milk chocolate → white chocolate → milk chocolate → dark chocolate
- Problems get more difficult when release/due dates need to be respected
- This is the equivalent to the *VRPTW (Vehicle Routing Problem with Time Windows)*

Xmas Shopping Hint

- W. Cook. In Pursuit of the Travelling Salesman. Princeton University Press, 2011
- Entertaining general science presentation of the TSP and related issues



Setup Times Constraints can be Included in Model



- Shown in Machine Gantt chart, enable display in `Layout` tab
- So far, only in CPO, not in CPSat model

Related Problem: Forbidden Transitions ✗

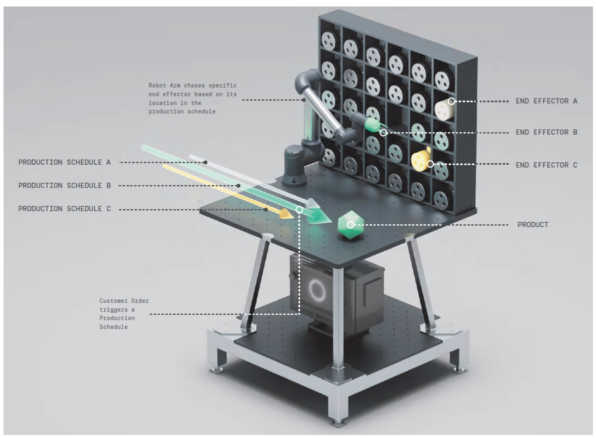
- For safety reasons, it may be forbidden to change from some product to some specific other products
- Contamination risk is considered too high
- Examples
  - In food production: Is this product peanut free?
  - In food production: Directly changing from dark to white chocolate is not allowed
  - In chemical plants: Contamination may lead to explosions
- These transitions are called *forbidden*, and must be avoided
- Careful, it is easy to paint yourself into a corner!

Dealing with Transportation Times

- Really two different problems
  - In one, the resources are in fixed locations, and we transport the jobs between the locations
  - In the other, the tasks are in fixed locations, and we transport the resources between them

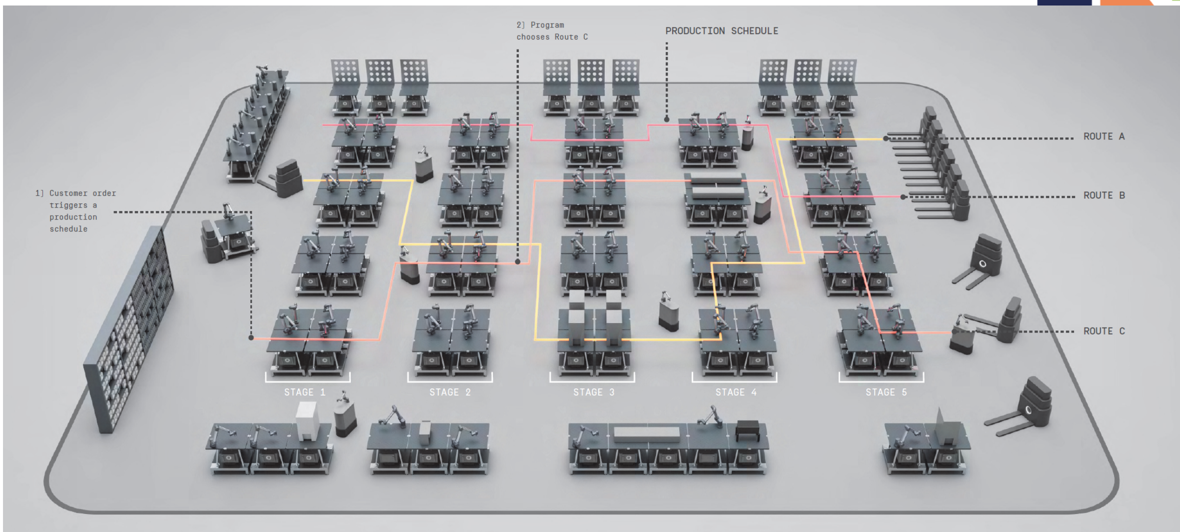
Transportation of Jobs

- Example from a project with J&J in Limerick
- Considering a *factory of the future* based on agile machines
- Robots that can be configured to perform many different tasks
- These robots may be inside one or more factories
- How to arrange them to minimize impact of transport on production



from J&J

Layout of Factor in Matrix Form



- Materials are transported between stations by moving robots
- Layout of factory determines delay caused by transport

Inclusion in Model (✓)

	Row	M0_0	M0_1	M0_2	M0_3	M0_4	M0_5	M0_6	M0_7	M0_8	M0_9	M1_0	M1_1	M1_2	M1_3	M1_4	M1_5	M1_6	M1_7	M1_8	M1_9	M2_0	M2_1	M2_2	M2_3	M2_4	M2_5	M2_6	M2_7	M2_8	M2_9	M3_0	M3_1	M3_2	M3_3	M3_4	M3_5	M3_6	M3_7	M3_8	M3_9
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2	M0_1	0	0	0	0	0	0	0	0	0	0	2	1	2	3	4	5	6	7	8	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	M0_2	0	0	0	0	0	0	0	0	0	0	3	2	1	2	3	4	5	6	7	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	M0_3	0	0	0	0	0	0	0	0	0	0	4	3	2	1	2	3	4	5	6	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	M0_4	0	0	0	0	0	0	0	0	0	0	5	4	3	2	1	2	3	4	5	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	M0_5	0	0	0	0	0	0	0	0	0	0	6	5	4	3	2	1	2	3	4	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	M0_6	0	0	0	0	0	0	0	0	0	0	7	6	5	4	3	2	1	2	3	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	M0_7	0	0	0	0	0	0	0	0	0	0	8	7	6	5	4	3	2	1	2	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	M0_8	0	0	0	0	0	0	0	0	0	0	9	8	7	6	5	4	3	2	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	M0_9	0	0	0	0	0	0	0	0	0	0	10	9	8	7	6	5	4	3	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
11	M1_0	1	2	3	4	5	6	7	8	9	10	0	0	0	0	0	0	0	0	0	0	1	2	3	4	5	6	7	8	9	10	0	0	0	0	0	0	0	0	0	0
12	M1_1	2	1	2	3	4	5	6	7	8	9	0	0	0	0	0	0	0	0	0	0	2	1	2	3	4	5	6	7	8	9	0	0	0	0	0	0	0	0	0	0
13	M1_2	3	2	1	2	3	4	5	6	7	8	0	0	0	0	0	0	0	0	0	0	3	2	1	2	3	4	5	6	7	8	0	0	0	0	0	0	0	0	0	0
14	M1_3	4	3	2	1	2	3	4	5	6	7	0	0	0	0	0	0	0	0	0	0	4	3	2	1	2	3	4	5	6	7	0	0	0	0	0	0	0	0	0	0
15	M1_4	5	4	3	2	1	2	3	4	5	6	0	0	0	0	0	0	0	0	0	0	5	4	3	2	1	2	3	4	5	6	0	0	0	0	0	0	0	0	0	0
16	M1_5	6	5	4	3	2	1	2	3	4	5	0	0	0	0	0	0	0	0	0	0	6	5	4	3	2	1	2	3	4	5	0	0	0	0	0	0	0	0	0	0
17	M1_6	7	6	5	4	3	2	1	2	3	4	0	0	0	0	0	0	0	0	0	0	7	6	5	4	3	2	1	2	3	4	0	0	0	0	0	0	0	0	0	0
18	M1_7	8	7	6	5	4	3	2	1	2	3	0	0	0	0	0	0	0	0	0	0	8	7	6	5	4	3	2	1	2	3	0	0	0	0	0	0	0	0	0	0
19	M1_8	9	8	7	6	5	4	3	2	1	2	0	0	0	0	0	0	0	0	0	0	9	8	7	6	5	4	3	2	1	2	0	0	0	0	0	0	0	0	0	0
20	M1_9	10	9	8	7	6	5	4	3	2	1	0	0	0	0	0	0	0	0	0	0	10	9	8	7	6	5	4	3	2	1	0	0	0	0	0	0	0	0	0	0
21	M2_0	0	0	0	0	0	0	0	0	0	0	1	2	3	4	5	6	7	8	9	10	0	0	0	0	0	0	0	0	0	0	1	2	3	4	5	6	7	8	9	10
22	M2_1	0	0	0	0	0	0	0	0	0	0	2	1	2	3	4	5	6	7	8	9	0	0	0	0	0	0	0	0	0	0	2	1	2	3	4	5	6	7	8	9
23	M2_2	0	0	0	0	0	0	0	0	0	0	3	2	1	2	3	4	5	6	7	8	0	0	0	0	0	0	0	0	0	0	3	2	1	2	3	4	5	6	7	8
24	M2_3	0	0	0	0	0	0	0	0	0	0	4	3	2	1	2	3	4	5	6	7	0	0	0	0	0	0	0	0	0	0	4	3	2	1	2	3	4	5	6	7
25	M2_4	0	0	0	0	0	0	0	0	0	0	5	4	3	2	1	2	3	4	5	6	0	0	0	0	0	0	0	0	0	0	5	4	3	2	1	2	3	4	5	6
26	M2_5	0	0	0	0	0	0	0	0	0	0	6	5	4	3	2	1	2	3	4	5	0	0	0	0	0	0	0	0	0	0	6	5	4	3	2	1	2	3	4	5
27	M2_6	0	0	0	0	0	0	0	0	0	0	7	6	5	4	3	2	1	2	3	4	0	0	0	0	0	0	0	0	0	0	7	6	5	4	3	2	1	2	3	4
28	M2_7	0	0	0	0	0	0	0	0	0	0	8	7	6	5	4	3	2	1	2	3	0	0	0	0	0	0	0	0	0	0	8	7	6	5	4	3	2	1	2	3
29	M2_8	0	0	0	0	0	0	0	0	0	0	9	8	7	6	5	4	3	2	1	2	0	0	0	0	0	0	0	0	0	0	9	8	7	6	5	4	3	2	1	2
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32	M3_1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	2	3	4	5	6	7	8	9	0	0	0	0	0	0	0	0	0	0
33	M3_2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	2	1	2	3	4	5	6	7	8	0	0	0	0	0	0	0	0	0	0
34	M3_3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	3	2	1	2	3	4	5	6	7	0	0	0	0	0	0	0	0	0	0
35	M3_4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	4	3	2	1	2	3	4	5	6	0	0	0	0	0	0	0	0	0	0
36	M3_5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	5	4	3	2	1	2	3	4	5	0	0	0	0	0	0	0	0	0	0
37	M3_6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	6	5	4	3	2	1	2	3	4	0	0	0	0	0	0	0	0	0	0
38	M3_7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	7	6	5	4	3	2	1	2	3	0	0	0	0	0	0	0	0	0	0
39	M3_8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	8	7	6	5	4	3	2	1	2	0	0	0	0	0	0	0	0	0	0
40	M3_9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	9	8	7	6	5	4	3	2	1	0	0	0	0	0	0	0	0	0	0
41	M4_0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	3	4	5	6	7	8	9	10	11	1	2	3	4	5	6	7	8	9	10
42	M4_1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	2	3	4	5	6	7	8	9	10	2	1	2	3	4	5	6	7	8	9
43	M4_2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	3	2	3	4	5	6	7	8	9	3	2	1	2	3	4	5	6	7	8
44	M4_3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	4	3	2	3	4	5	6	7	8	4	3	2	1	2	3	4	5	6	7
45	M4_4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	5	4	3	2	3	4	5	6	7	5	4	3	2	1	2	3	4	5	6
46	M4_5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	6	5	4	3	2	3	4	5	6	6	5	4	3	2	1	2	3	4	5
47	M4_6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	7	6	5	4	3	2	3	4	5	7	6	5	4	3	2	1	2	3	4
48	M4_7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	8	7	6	5	4	3	2	3	4	8	7	6	5	4	3	2	1	2	3
49	M4_8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	9	8	7	6	5	4	3	2	3	9	8	7	6	5	4	3	2	1	2
50	M4_9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	10	9	8	7	6	5	4	3	2	10	9	8	7	6	5	4	3	2	1

- Add location attribute to each resource
- Include transport time as element in temporal constraints

More Complex Variant ✗

- Schedule the moving robots as well
- Assume that an empty robot travels much faster than a loaded one
- We can treat the robots as a machine choice resource for the transportation tasks

Even More Complex Variant ✗

- Schedule the moving robots as well
- They move at the same speed empty and loaded
- We can to bring them from the end of one transport task to the start of the next one
- This is a vehicle routine problem
- In some industries, this is the harder problem than scheduling the plant itself
  - Torpedo scheduling in steel plant: rail cars holding molten steel, quantities limited

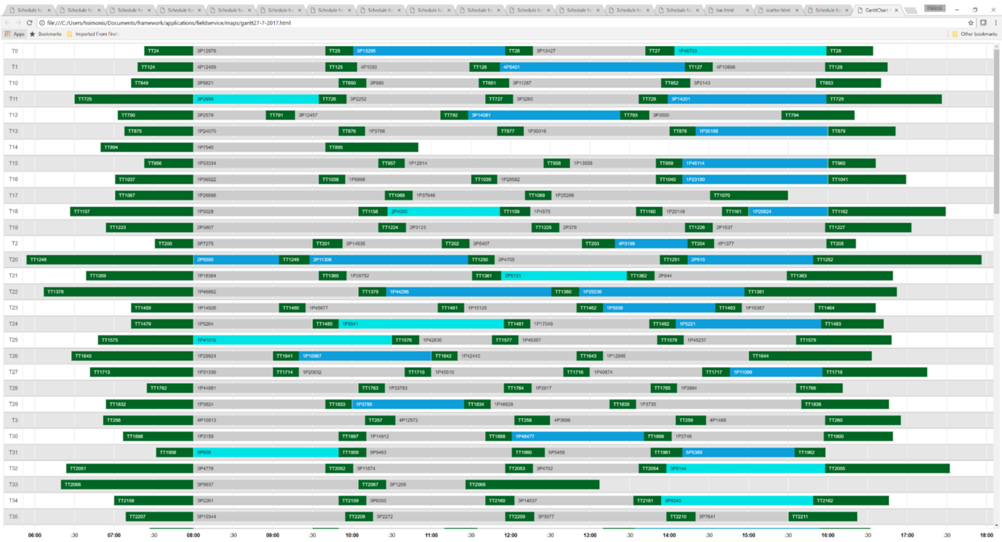
Torpedo Scheduling (CP 2016 Challenge)

(from ACP Website http://cp2016.a4cp.org/program/acp-challenge/)

Scheduling Service Visits ✗

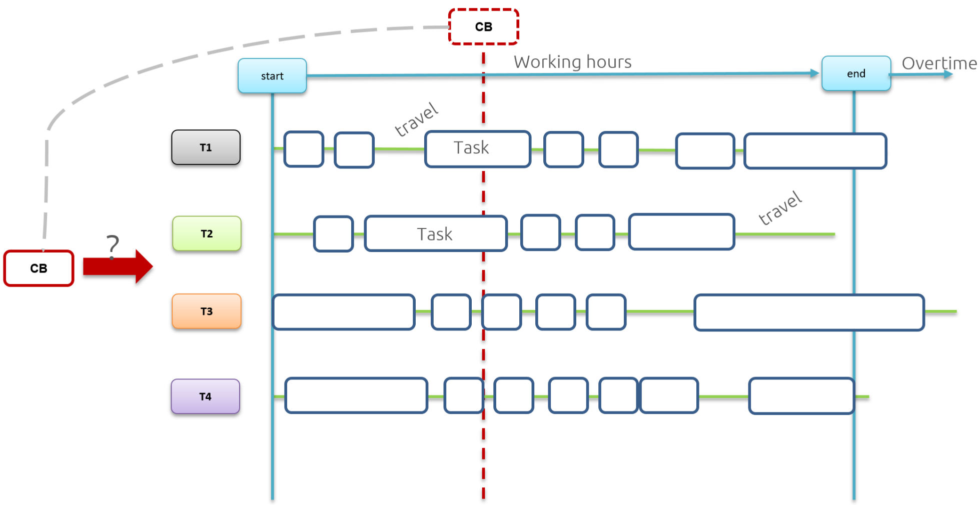
- Based on a project with UTRC-I, UTRC, OTIS
- Schedule visits to maintain equipment installed in customer premises
- Resources are the service engineers
- They have to travel between locations and perform work there
- The tasks are the maintenance operations required to keep equipment working
- Also called *Traveling Repair-person Problem*

Planning Maintenance Visits for Service Personnel



- Include single day trips, multi-day tours
- Most of the time spent at customer locations

Re-scheduling Problem



- How to react when a customer is trapped in an elevator
- All your engineers are on service calls
- *Who you gonna call?*

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- This will be described in more detail in a new course
- AI Fundamentals: Skill Development Program on Transportation Optimization
- Arriving in 2025 at this location

Summary

- We presented some more advanced topics
  - Sequence dependent setup
  - Transportation time
- Not available in every solver
- Useful concepts when dealing with specific scheduling problems
- Leading to another *Skills Development Program*